



Marathwada Mitra Mandal's
College of Engineering
An Autonomous Institute



**DEPARTMENT OF COMPUTER
ENGINEERING**

WELCOMES
For
Engineering Design and Innovation
Mini Project (Semester V)
Implementation Framework Presentation

09th July 2026

Agenda

- Objectives and Key outcomes
- Foundation Courses
- Connecting Foundation Courses to the Project Theme
- Sample Project
- Implementation Plan
- Conduction Plan
- Assessment and Required Evidence
- Budget

Objectives and Key Outcomes

Objectives

- To Integrate knowledge gained from **foundational courses** studied in previous semesters.
- Provides an opportunity to apply theoretical concepts in a practical project.
- Strengthens understanding of core subjects through hands-on implementation.
- Encourages problem-solving, innovation, and teamwork.
- Bridges the gap between academic learning and real-world applications.

Key Outcomes

- Reinforces concepts learned in earlier semesters.
- Develops practical skills using relevant tools and technologies.
- Improves project planning, design, implementation, and testing abilities.
- Enhances communication, documentation, and presentation skills.
- Prepares students for major projects, internships, and industry challenges.

Foundation Courses

First-year/common foundational courses:

1. Fundamentals of Programming Languages
2. Engineering Exploration Lab I & II

Second-year department core courses:

1. Data Structures
2. Object Oriented Programming
3. Digital Electronics and Computer Organization
4. Database Management Systems
5. Principles of Programming Languages
6. Software Engineering
7. Web Technology
8. Operating System Workshop

Connecting Foundation Courses to the Project Theme

- Foundation courses from previous semesters provide the theoretical knowledge required for the mini project.
- The project theme integrates concepts from multiple subjects to solve a practical problem.
- Students apply knowledge from core courses such as programming, databases, networking, electronics, mathematics, or domain-specific subjects (as applicable).
- The project demonstrates how individual course concepts work together in a real-world application.
- This integration strengthens conceptual understanding and develops practical, interdisciplinary skills

Connecting Foundation Courses to the Project Theme

Foundational Course	Contribution to Project
Programming Fundamentals	Logic Building and Problem Solving
Data Structures	Efficient Algorithms
Object-Oriented Programming	Modular Software Design
DBMS	Data Storage & Management
Software Engineering	SDLC, Documentation & Testing
Digital Electronics & CO	System-Level Understanding
Web Technology	User Interface Development
Principles of Programming Languages	Application Development

Sample Project:

Campus Event and Activity Management Portal

Event registration, approval workflow, participant management, access control, API-based architecture.

Foundational Courses Integrated

- Object-Oriented Programming
- DBMS
- Web Technologies
- Software Engineering

Core Concepts to be Consolidated

- Event lifecycle
- Approval workflow
- User authentication
- Database normalization
- REST APIs
- Form validation
- Testing strategies

Sample Project:

Campus Event and Activity Management Portal

Event registration, approval workflow, participant management, access control, API-based architecture.

Foundational Courses Integrated :

Course	Concepts	Expected Outcomes / KPI's
Object-Oriented Programming	User Objects, Event objects, Inheritance, Event Triggers and Function Calls	System/Project Architecture Diagram, Functions, Object Interactions - Code Duplication Rate
DBMS	ER diagram,Joins, Views, Normalization,JDBC, ACID properties, Concurrency control, Performance check, cursors and Triggers	3NF Expectation, Index Utilization, Deadlock and Lock Wait Time, Zero redundant join rewrites
Web Technologies	Server side application,role based access control,form validation,UI design, REST/SOAP API	Response Latency of Server, Uptime & Resource Efficiency, Form Validations
Data Structures	Lists,array,Searching/Sorting algorithms,	Time and space complexity
Software Engineering	SRS,Testing strategies,Version Control	Requirements Traceability Matrix , Merge Requests, Commit and Deployment Frequency

Sample Project:

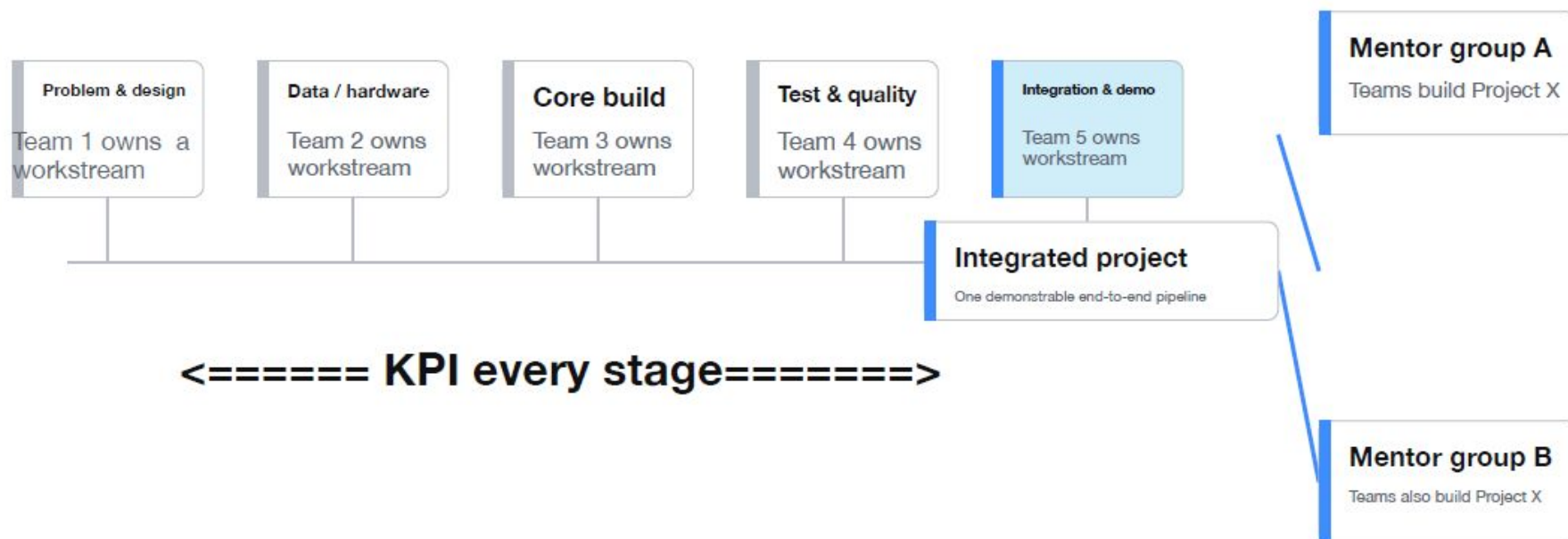
Campus Event and Activity Management Portal

Event registration, approval workflow, participant management, access control, API-based architecture.

Foundational Courses Integrated :

Course	Concepts	Expected Outcomes / KPI's
Object-Oriented Programming	User Objects, Event objects, Inheritance, Event Triggers and Function Calls	1. 100% automation 2. The portal must handle up to 50-100 concurrent registration requests per minute without system degradation 3. Less than 50ms for API-driven event queries and search listings. 4. 99.9% Server Uptime 5. 0% unauthorized access incidents. 6. Measure of Space and Time Complexity
DBMS	ER diagram, Joins, Views, Normalization (3NF), JDBC, ACID properties, Concurrency control, Performance check, cursors and Triggers	
Web Technologies	Server side application,role based access control,form validation,UI design, REST/SOAP API	
Data Structures	Lists,array,Searching/Sorting algorithms,Time and space complexity	
Software Engineering	SRS,Testing strategies,Version Control	

Learning Core Engineering knowledge



Learning rule: ownership may be specialized, but reviews, interface contracts, demos and retrospectives must expose every team to the full pipeline.

Every team reports the same five learning KPIs

Project-specific measures vary; these common KPIs allow meaningful discussion across all five teams and across mentor groups executing the same project.

01

System Design, Feature design

Core engineering concepts are visible in design choices, code, tests and explanations.

02

Continuous integration

Frequent merges, automated builds, integration success and recovery from failed builds.

03

Instrumentation & observability

Skill creation, Logs, metrics, traces, test points or measurements make system behaviour visible.

04

Collaborative delivery

Dependencies, peer reviews, interface agreements and shared fixes move toward closure.

05

Leadership & feedback

Rotation is completed, decisions are recorded and mentor or HOD feedback closes on time.

<===== KPI every stage=====>

Within each mentor group

Discuss the five-team pattern: where is learning strong, where is evidence weak, and which dependency needs joint action?

Across teams executing the same project

Co-mentors compare the same KPI definitions, identify recurring causes and share practices—without ranking teams out of context.

Discussion rule: compare trends and evidence first; use project-specific KPIs as a second layer.

Student team weekly report — one-page template

SUBMIT THU • 10:00 PM IST



PROJECT / FEATURE

[Project name] • [Feature / sub-project]

TEAM & MENTOR

[Team ID] • [Faculty mentor]

WEEK /Team/LEADER

[Week ending] • [Student name]

WORK COMPLETED

3–5 outcome-led bullets.

Include links or evidence: commit, test result, design, dataset, demo.

CHALLENGES / RISKS

3-5 BULLETS==>State the issue, its impact, what was tried, and current severity: Red / Amber / Green.

HELP NEEDED

Ask ONE precise question. Name the decision, resource, access or technical guidance required—and by when.

INTEGRATION & COLLABORATION

Interfaces used • teams consulted • dependencies resolved/open • what the team learned about the end-to-end pipeline.

COMMON KPI EVIDENCE

Foundational behaviour [CPU Usage/Memory Userd/Latency of execution]• continuous integration • instrumentation • collaborative delivery • leadership and feedback.

NEXT WEEK COMMITMENTS

Owner + deliverable + due date for each commitment. Carry forward unresolved mentor actions.

MENTOR RESPONSE

Decision / guidance / owner / target date.

Leader submits; all four.

Faculty mentor weekly consolidation template

5 TEAM INPUTS → 1 MENTOR VIEW

PROJECT / MENTOR GROUP

[Project name] • [Mentor] • [Week ending]

OVERALL STATUS

RAG: ☐ Red ☐ Amber ☐ Green • Common KPI trend: [Improving / Stable / Declining]

Team / workstream	Outcome this week	Common KPI signal	Risk / dependency	Mentor action / owner
Team 1 / [feature]				
Team 2 / [feature]				
Team 3 / [feature]				
Team 4 / [feature]				
Team 5 / [feature]				

CROSS-TEAM KPI SYNTHESIS

What common KPI pattern appears across the five teams? What cause or shared action explains it?

ESCALATIONS / HELP FROM HOD

Decision, resource, policy, lab access or external support needed—with owner and required date.

NEXT-WEEK PRIORITIES

Top 3 integrated outcomes, critical review points and any change to scope.

SAME-PROJECT KPI DISCUSSION

Co-mentor(s), KPI trend comparison, shared practice or intervention, joint owner and review date.

Implementation Plan

- Mapping of Mini Project course with Engineering Design and Innovation(EDI) Lab
- [Syllabus Link](#) [Rubrics](#)

Week	Activity	Evaluation
Week 1	Orientation & Team Formation	Practical Continuous Assessment Sheet (CAS)
Week 2-3	Problem identification	
Week 4	Project Synopsis Submission (Theme Approval)	REVIEW 1
Week 5-7	Design & implementation	Practical CAS
Week 8-9	Subsystem Implementation	REVIEW 2
Week 10-11	Integration and Validation	Practical CAS
Week 12	Prototype Demonstration	Practical CAS
Week 13-14	Testing & documentation	Practical CAS
Week 15	Final demo & individual viva	REVIEW 3

Conduction Plan

Sr.No	Class	Batch	Identified Faculty
1	TY I	Batch A	1. Dr. Geetha Chillarge 2. Dr. Smita.Chaudhari 3. Ms. Vandana Rupnar 4. Mr. Rudragouda Patil 5. Ms. Swati Sardeshmukh 6. Mr. Mahesh kandekar
2		Batch B	
3		Batch C	
4	TY II	Batch A	
5		Batch B	
6		Batch C	

EDI Slots are mentioned in Regular Time table

Assessment and Required Evidence

Assessment:

- Assessment (25 TW)= Practical CAS (20)+ Review 1 (10)+ Review 2 (10)+ Review 3 (10) =50 convert it to 25.

Evidences

- Requirements document
- System architecture diagram
- Database schema or data model
- API specification
- Algorithm or business logic
- Test cases with results
- Debugging log
- Git history

Thank You